Expt: 4
Date:

Setup and configuration of local area Network using a switch and Ethernet cables.

* Aim:

To setup and configure a local area Network using a switch and Ethernet cables, assign IP address and verify communication between computers.

* LAN:

A Local Area Network connects computers and devices within a small area such as a home, office, or school. It enables users to shares files, printers, Internet access, and other resources. A switch connects all devices and allows fast and secure communication.

* Step to set Up a LAN:

1] Plan and Design a appropriate network topology taking into account Network requirement

and equipment location.

2] Take 4 computers a Switch with 8, 16, or 24 ports which is sufficient for Networks of these sizes and 4 Ether net cables.

3] Connect the computer to Network Switch via an Ethernet cables, which is a simple as Plugging one end of the Ethernet Cables into the Computer and other end into the Network Switch.

4] Assign Ip address to your PC's.

i) Log on the Computer as Administration or as owner.

ii) Click Network and Internet connection.

iii) Right click local area connection / Ethernet -> Go to properties -> select Internet Protocol. click on properties select use the following ip address option and assign ip address.

Similary assign Ip address to all the PCs connected to switch

PC1 - IP address : 10.1.1.1, Subnet mask 255.0.0.0

PC2 - IP address : 10.1.1.2, Subnet mask 255.0.0.0

PC3 - IP address - 10.1.1.3, Subnet mask 233.0.0.0

PC4 - IP address - 10.1.1.4, Subnet mask 233.0.0.0

5] Configure the switch with the IP address 10.1.1.3.

6] Check the connectivity between switch and other machine by using ping command in the command prompt of the device.

7] Select a folder → go to properties → click Sharing tab → share it with everyone on the same lan.

8] Try to access the shared folder from other computer of the Networks.

* Result:

A LAN was successfully configured using a switch and Ethernet cables. IP addresses were assignment correctly, communication was verified using the ping command and file sharing was successfully demonstrated between computers on the LAN.